

**Problem No: 01****Problem Name:** Socket Client-Server Communication in Python**Objective:**

- To implement a TCP client-server communication system using Python.
- To understand the basic concepts of socket programming and network communication.

**Theory:**

- **Socket:** Endpoint for sending or receiving data across a network.
- **TCP/IP:** Protocol that ensures reliable communication between two systems.
- **Client-Server Model:**
  - **Server:** Waits for connections from clients and responds.
  - **Client:** Initiates connection to send/receive data.
- **Port Number:** Identifier for a specific process/application on a machine (Port 1002 used here).

**Key Components:**

- Python socket library
- TCP protocol for reliable communication
- Server and client programs
- Localhost (127.0.0.1) as host

**Application:**

- Chat applications
- Online gaming communication
- File transfer programs
- Network services

**Implementation in Python:****Server Code (server1002.py)**

```
import socket
server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server_socket.bind(('localhost', 1002))
server_socket.listen(1)
print("Server is listening on port 1002...")
conn, addr = server_socket.accept()
print(f"Connected by {addr}")
while True:
    data = conn.recv(1024)
    if not data:
        break
    conn.sendall(data)
conn.close()
server_socket.close()
```

**Client Code (client1002.py)**

```
import socket
client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client_socket.connect(('localhost', 1002))
```

```
while True:
    message = input("chithi deo: ")
    if message.lower() == 'monowar':
        break
    client_socket.sendall(message.encode())
    data = client_socket.recv(1024)
    print(f'Received from server: {data.decode()}')
client_socket.close()
```

**Result / Discussion:**

- Client successfully connects to the server.
- Messages sent by the client are echoed back by the server.
- Communication terminates correctly when the client exits.

**Sample Output:****Server Console:**

Server is listening on port 1002...  
Connected by ('127.0.0.1', 65129)

**Client Console:**

chithi deo: lab  
Received from server: lab  
chithi deo: monowar

**Discussion:**

In this lab, we implemented a TCP client-server system in Python. The client successfully connected to the server, sent messages, and received responses. This demonstrates how TCP ensures reliable communication and how the client-server model works.

**Conclusion:**

The lab helped us understand socket programming and the client-server architecture. Messages were transmitted and received correctly, showing successful implementation of network communication using Python.